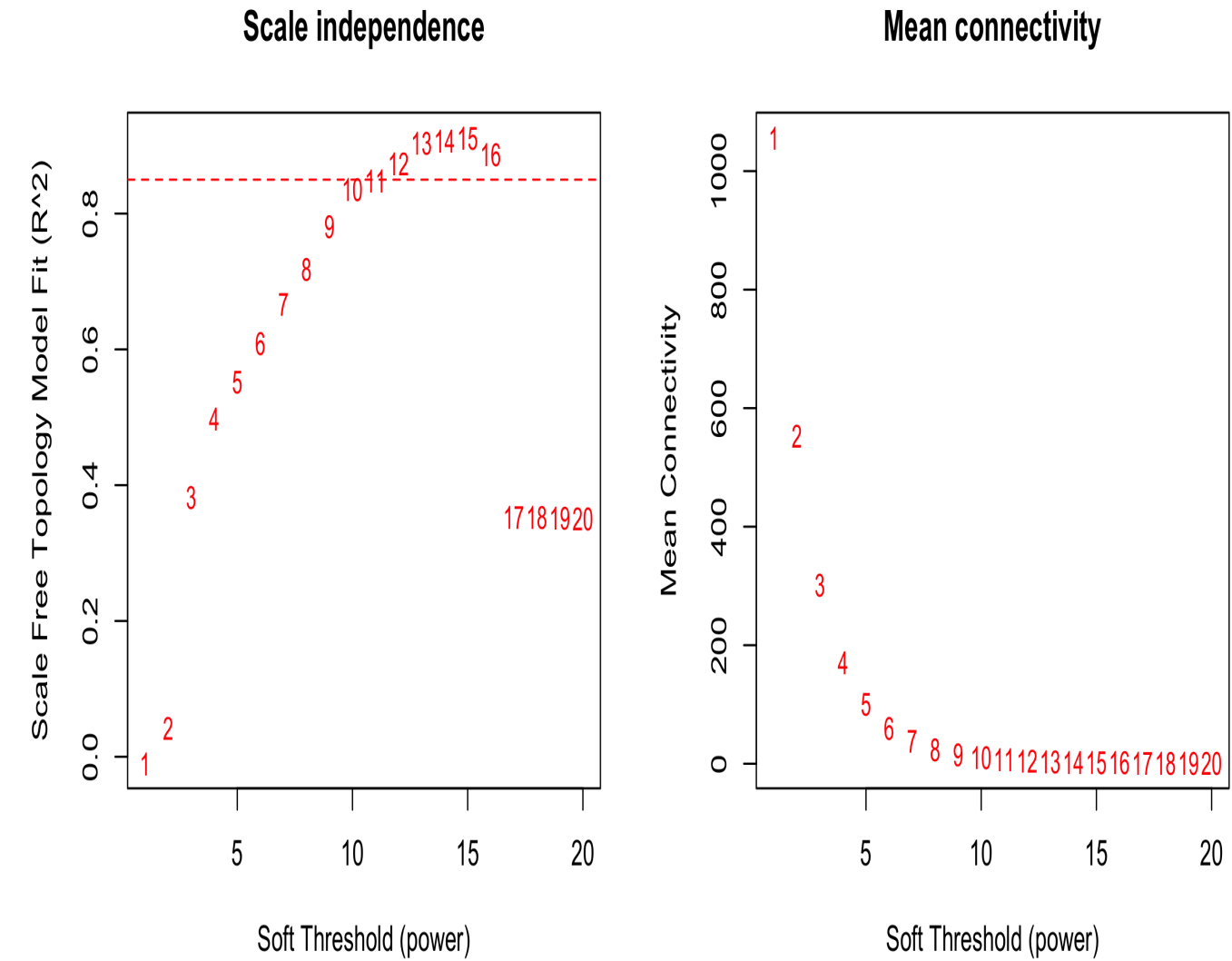


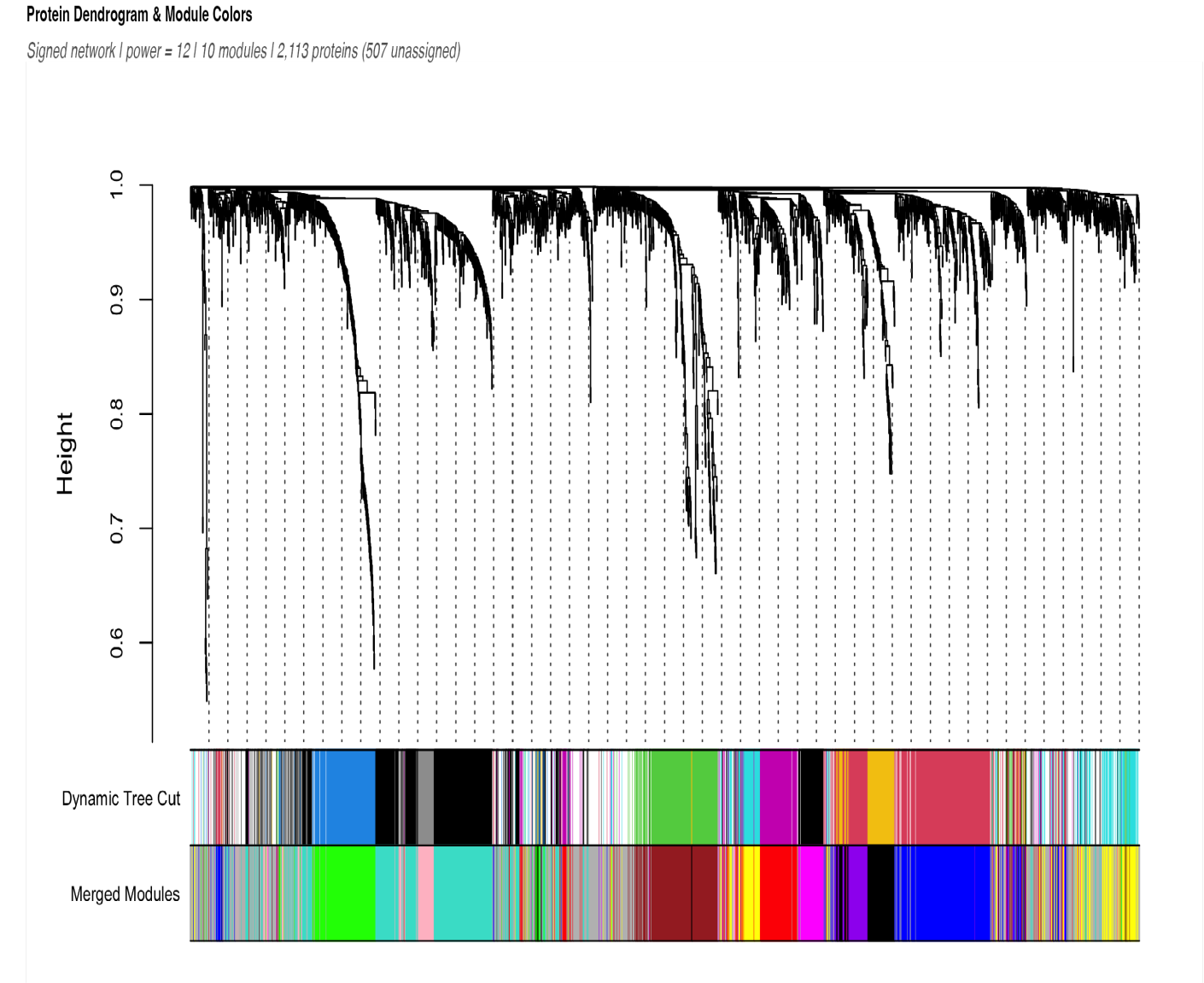
S7a – WGCNA Quality Control

A: Soft threshold | B: Dendrogram | C: Compartment enrichment | D: Module–trait heatmap | E: Bicor sensitivity

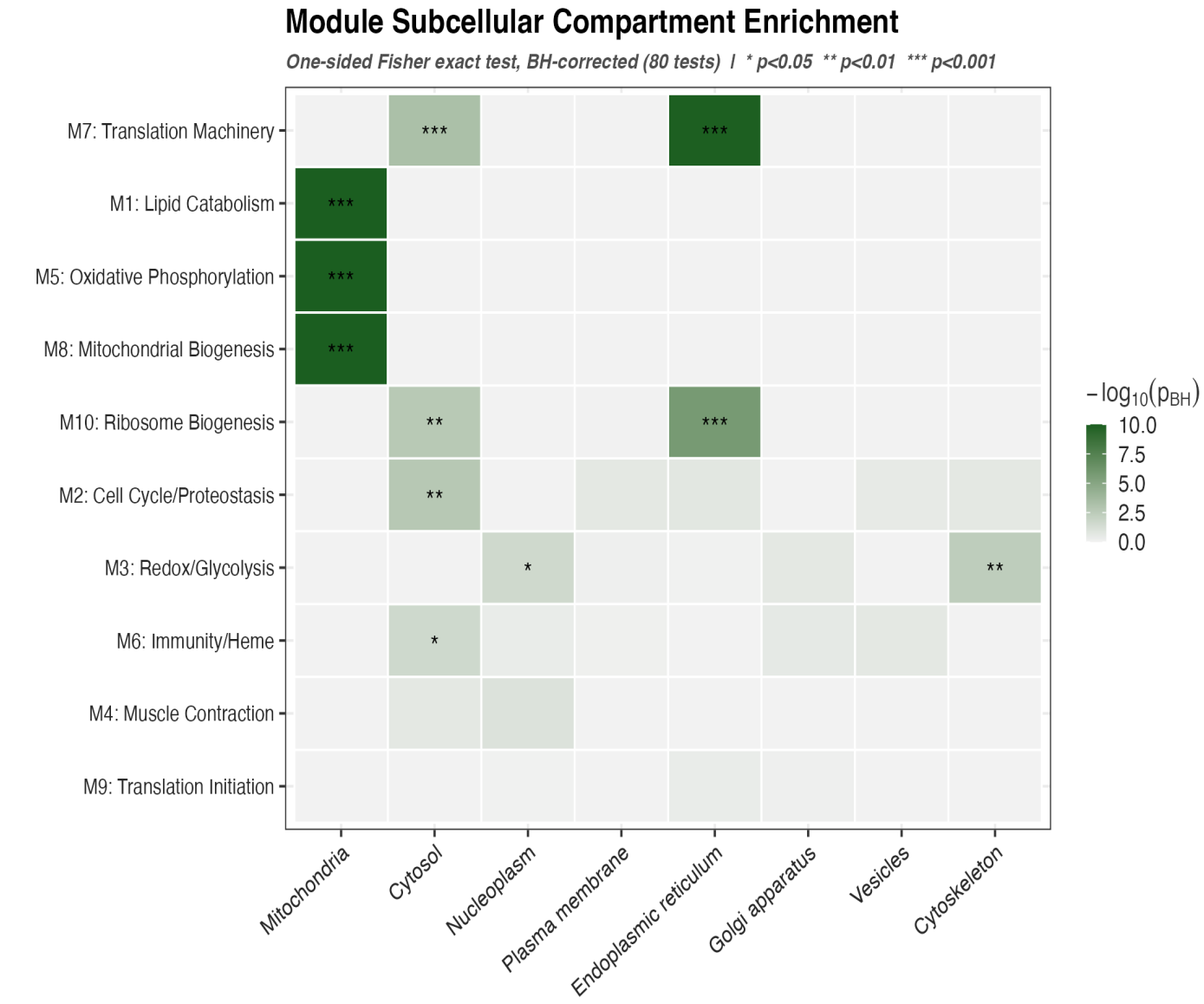
A



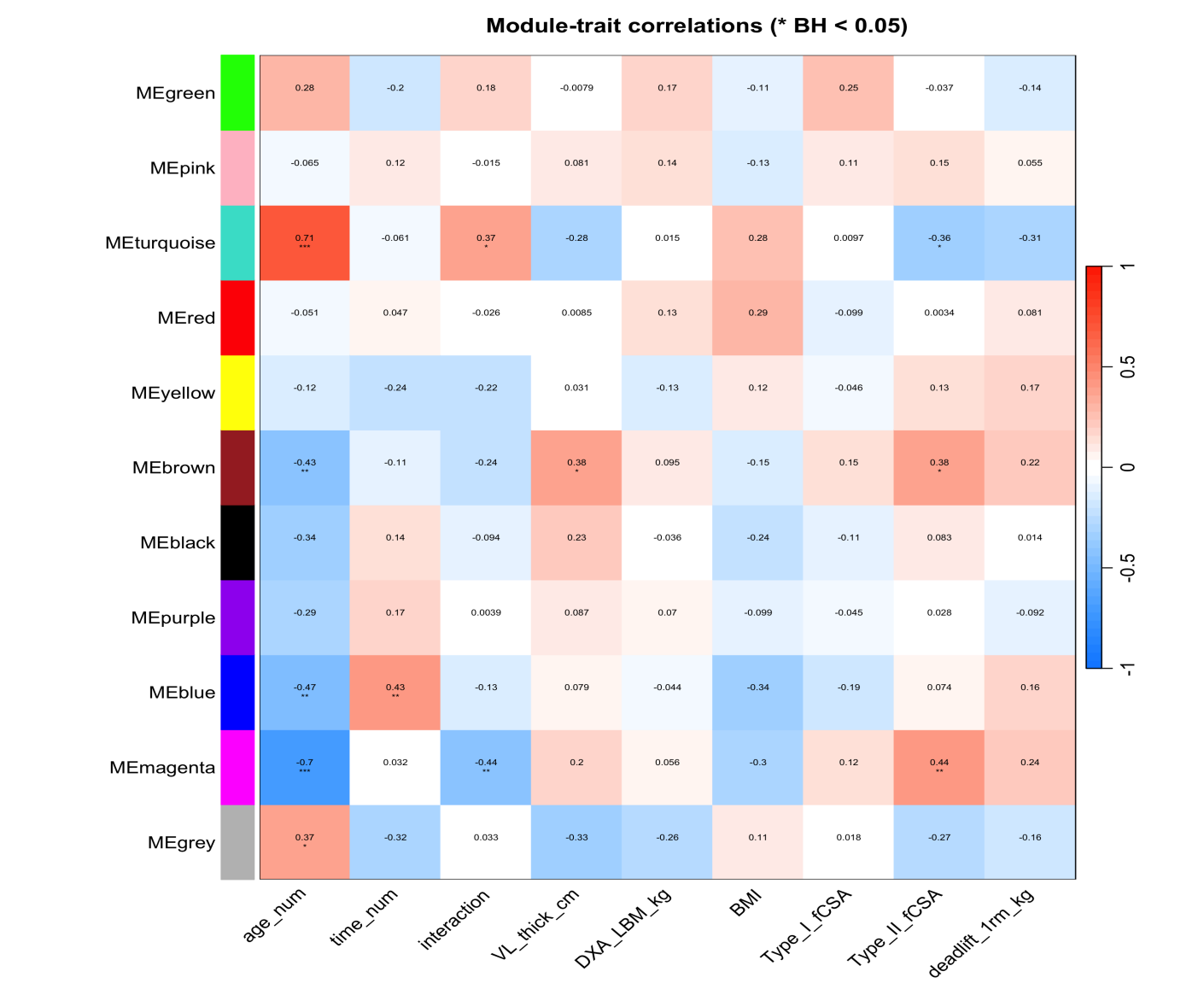
B



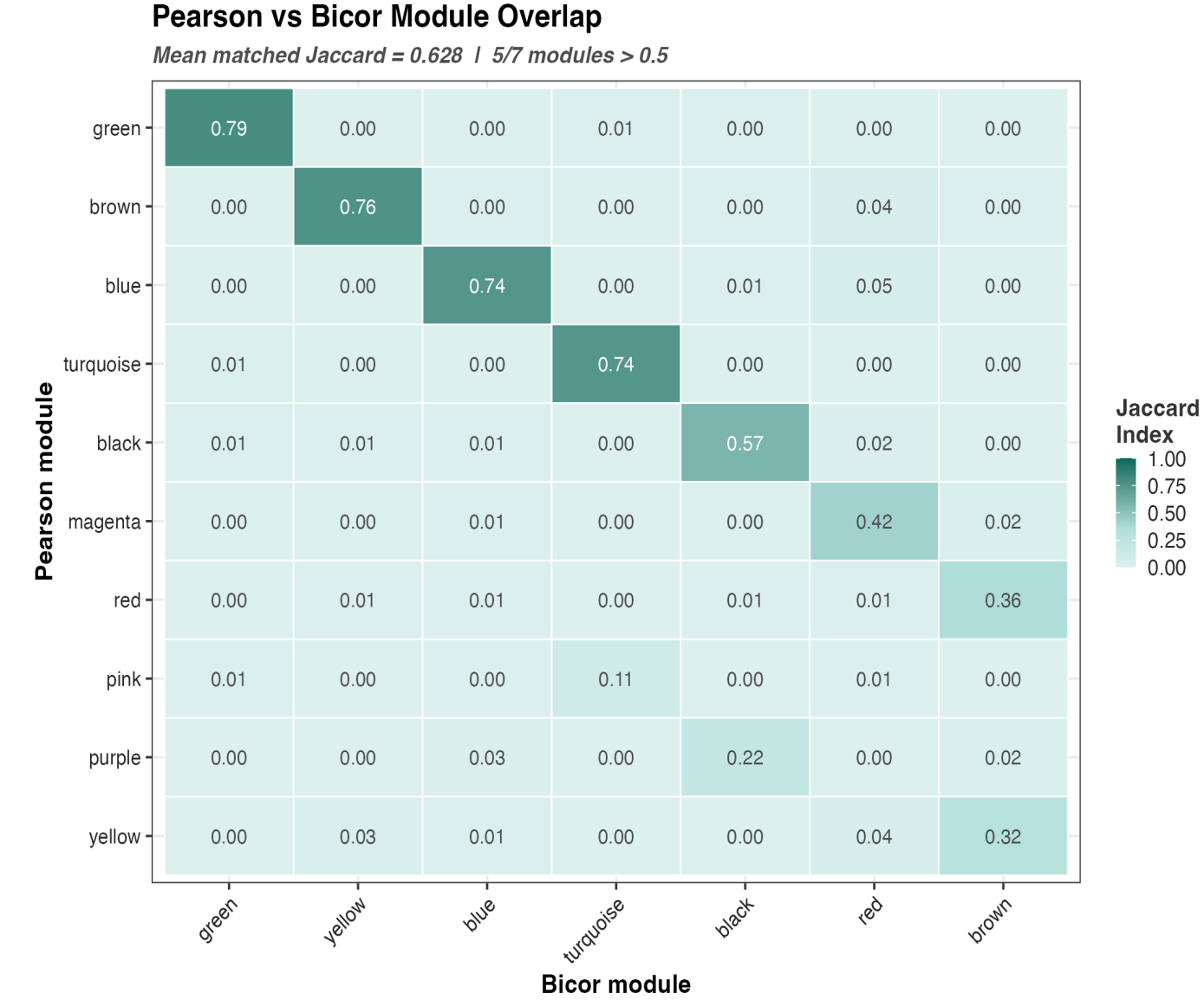
C



D



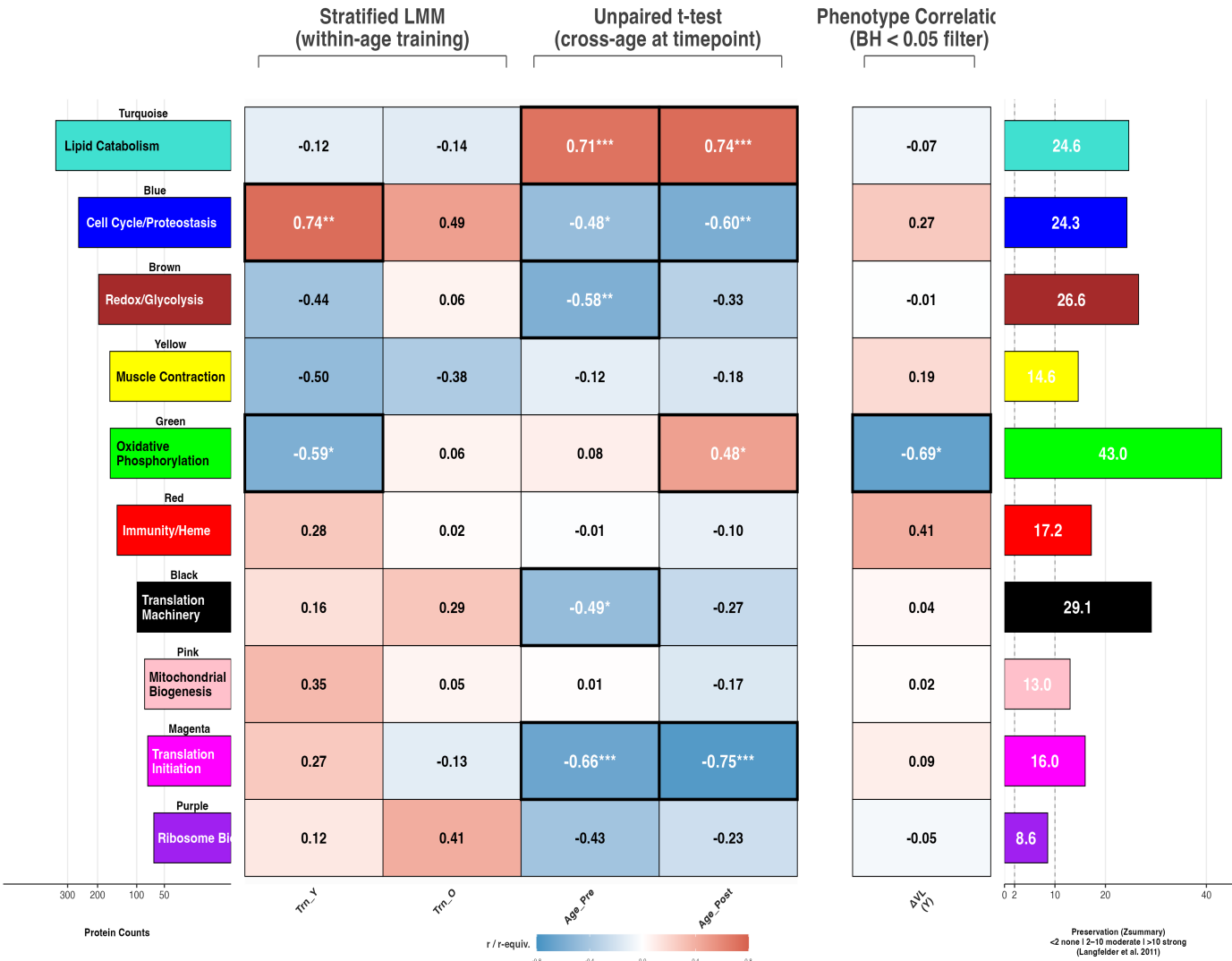
E



A

Age-Stratified Analysis — Training, Aging, & Phenotype Associations

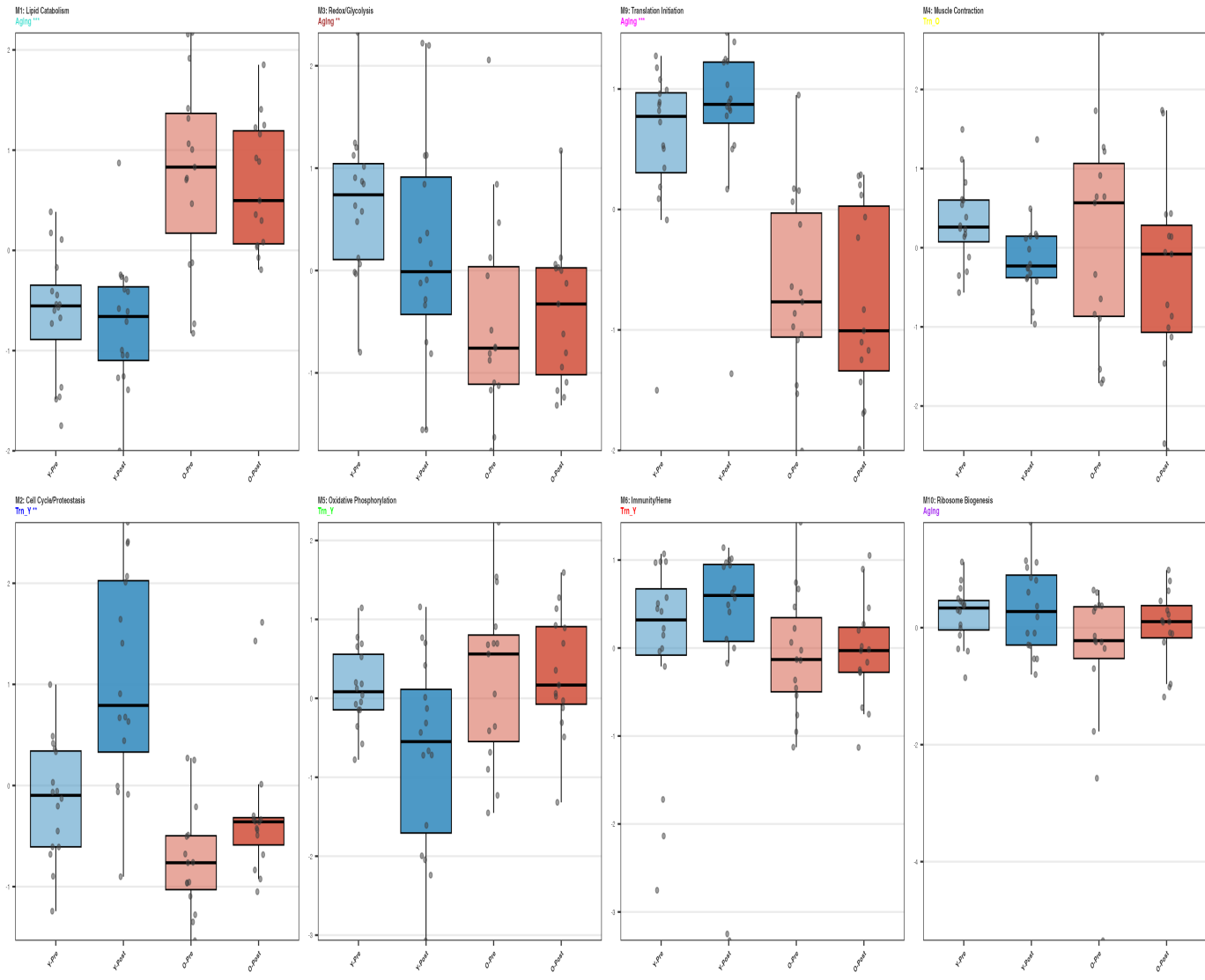
10 modules | LMM: eigengene ~ time + (1|subject), BH per stratum | Aging: unpaired t-test, BH per timepoint | Phenotype: Pearson r, BH per column
* p<0.05 ** p<0.01 *** p<0.001 † p<0.1 | Black border = BH < 0.05



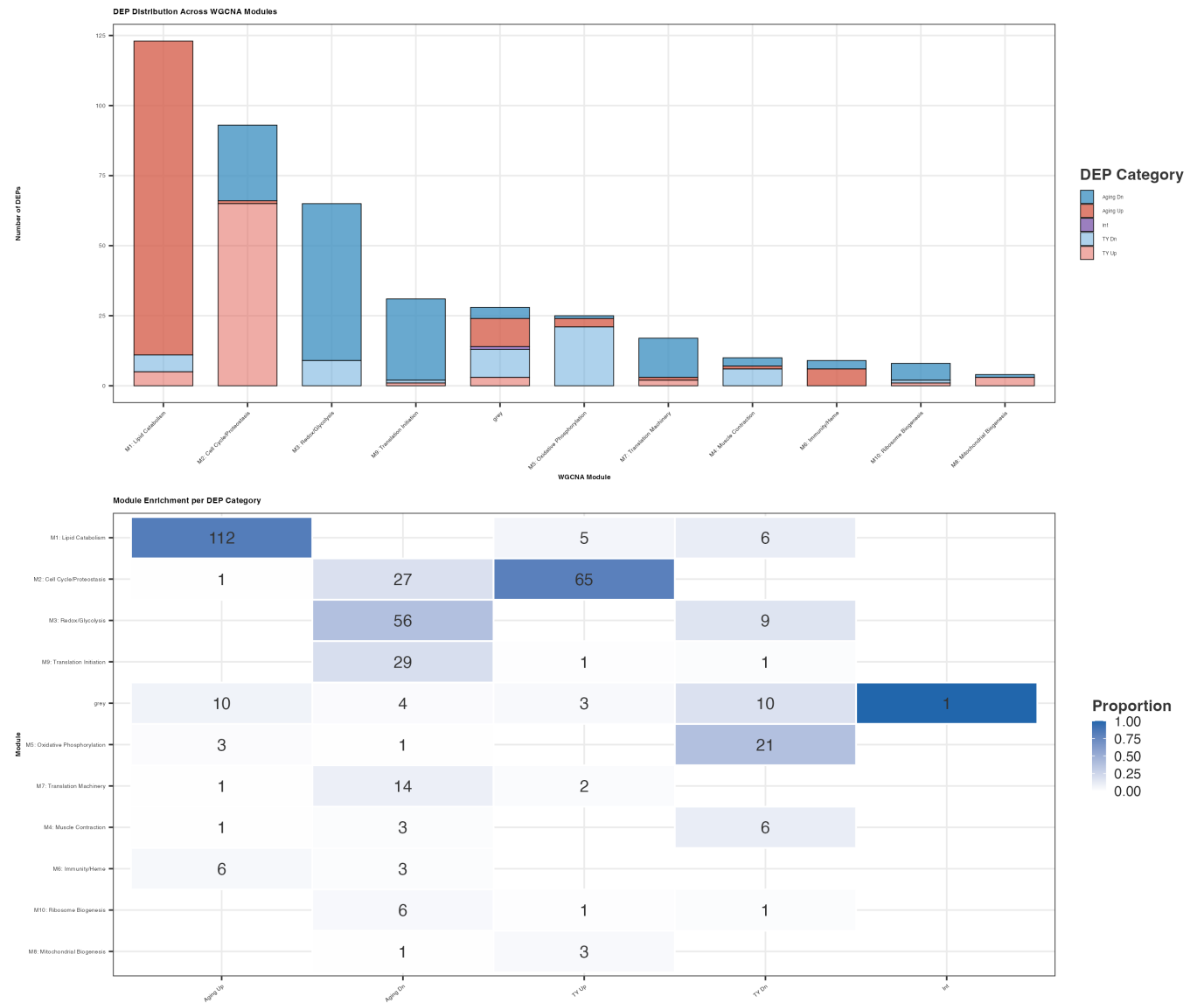
B

Eigengene Exemplars — 8 modules (BH<0.05 or 10<0.4)

Z-scored module eigengenes by Group x Time Aging ~ time | Aging ~ time | Aging ~ time | Aging ~ time | Aging ~ time | Aging ~ time | Aging ~ time | Aging ~ time



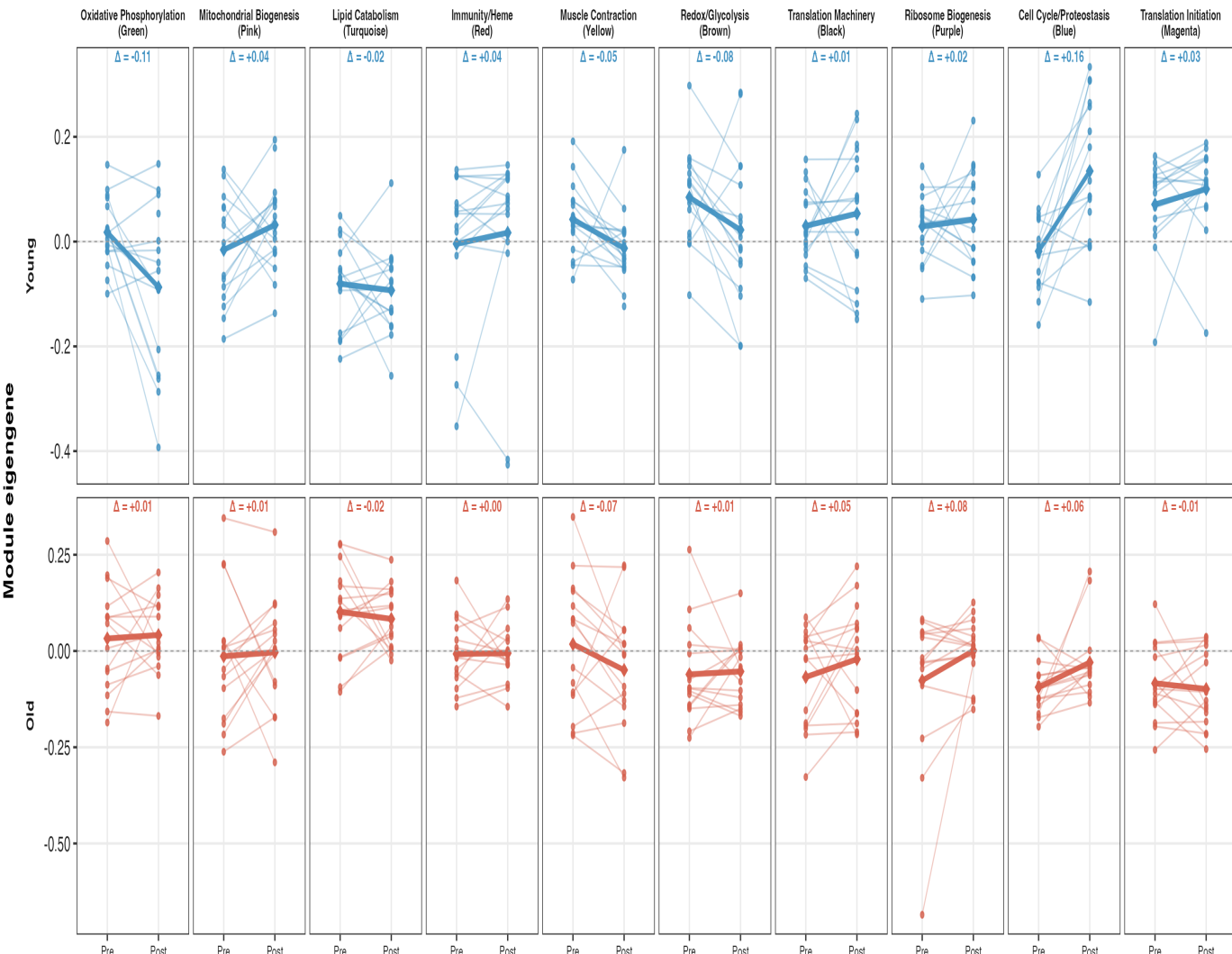
C



D

Module Eigengene Trajectories (Pre → Post)

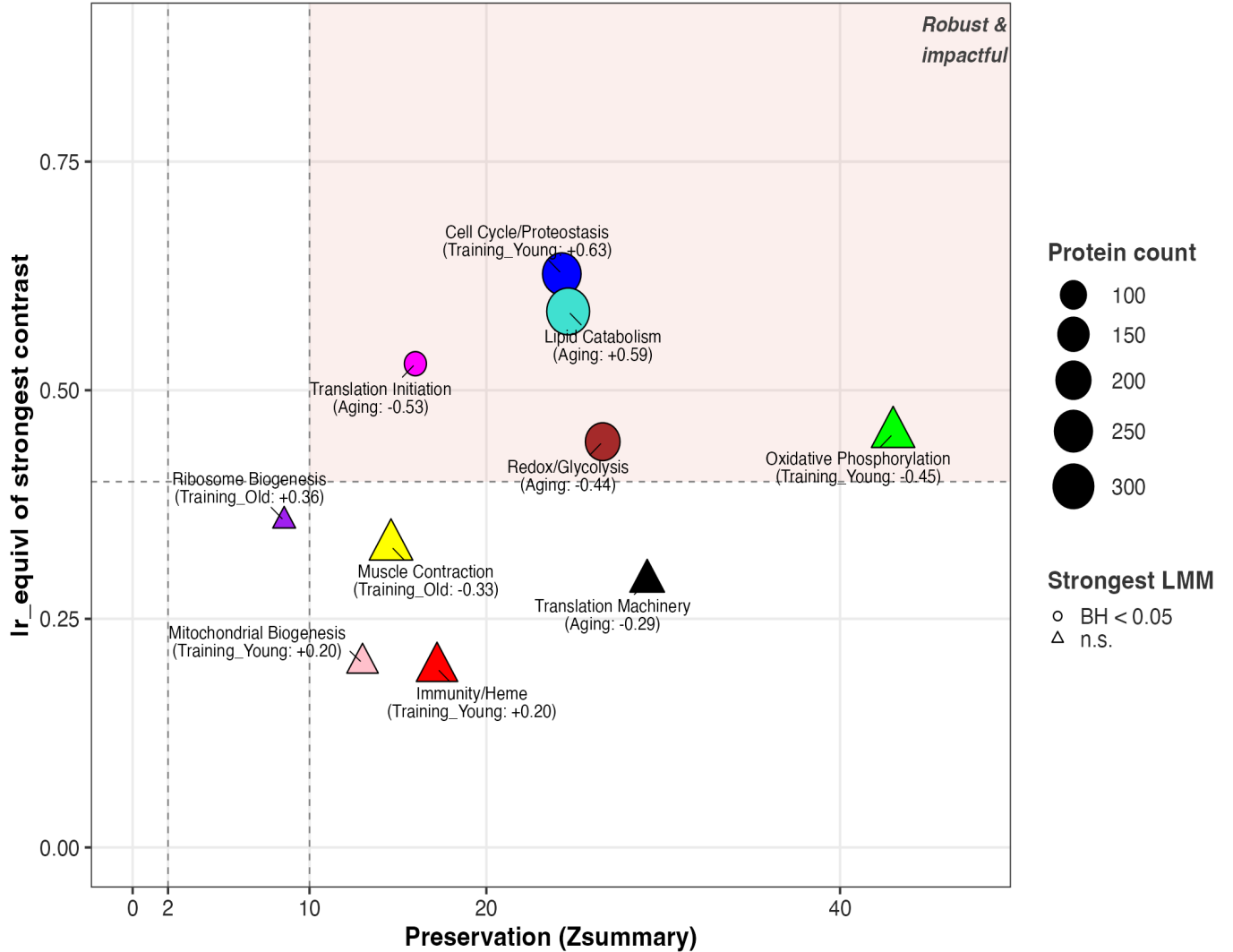
Per-subject ME paths | bold line = group mean | diamonds = group means



E

Module Preservation vs. Strongest Effect Size

Zsummary (Langfelder 2011) | Ir_equiv from LMM contrast audit



S7c – Module Triptychs (Key Modules)

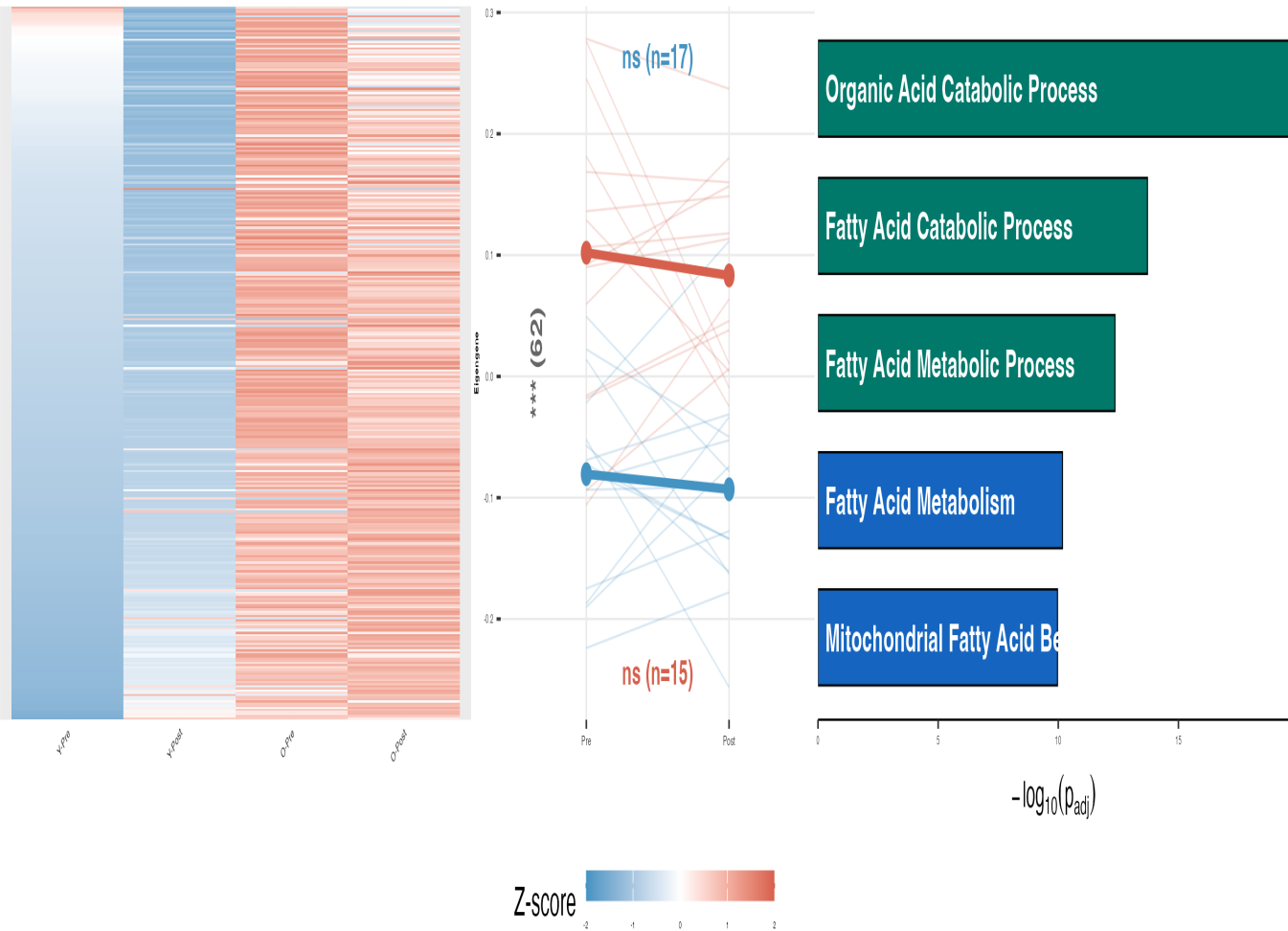
A: Turquoise (Lipid) | B: Blue (Cell Cycle) | C: Green (OxPhos) | D: Brown (Redox) | E: Magenta (Translation)

A

M1: Lipid Catabolism

Aging: $r = +0.59$

M1: Lipid Catabolism (n=334)



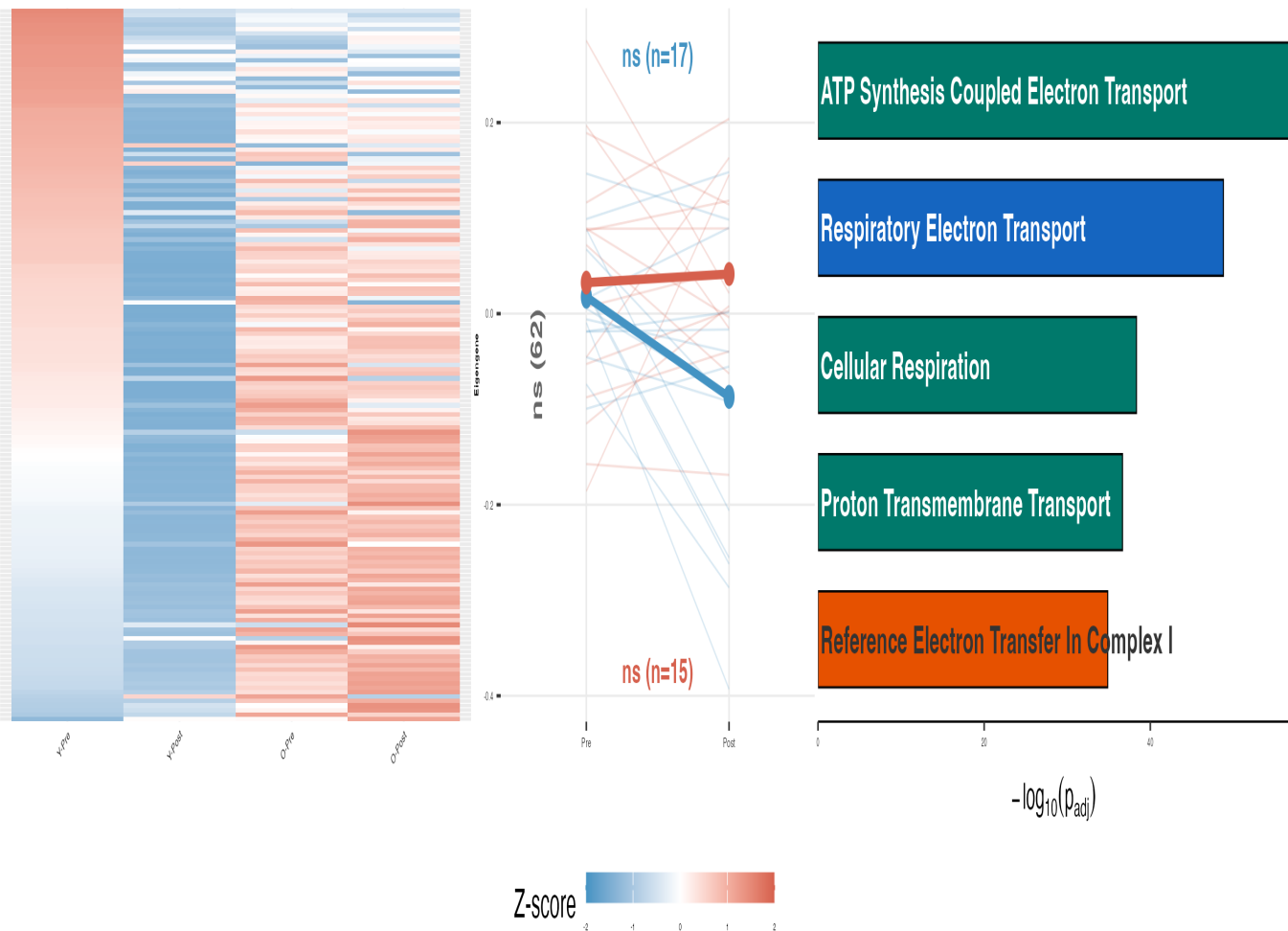
*FDR < 0.5 **FDR < 0.01 ***FDR < 0.001 (all LMM Bonferroni)

C

M5: Oxidative Phosphorylation

Aging: $r = +0.45$

M5: Oxidative Phosphorylation (n=159)



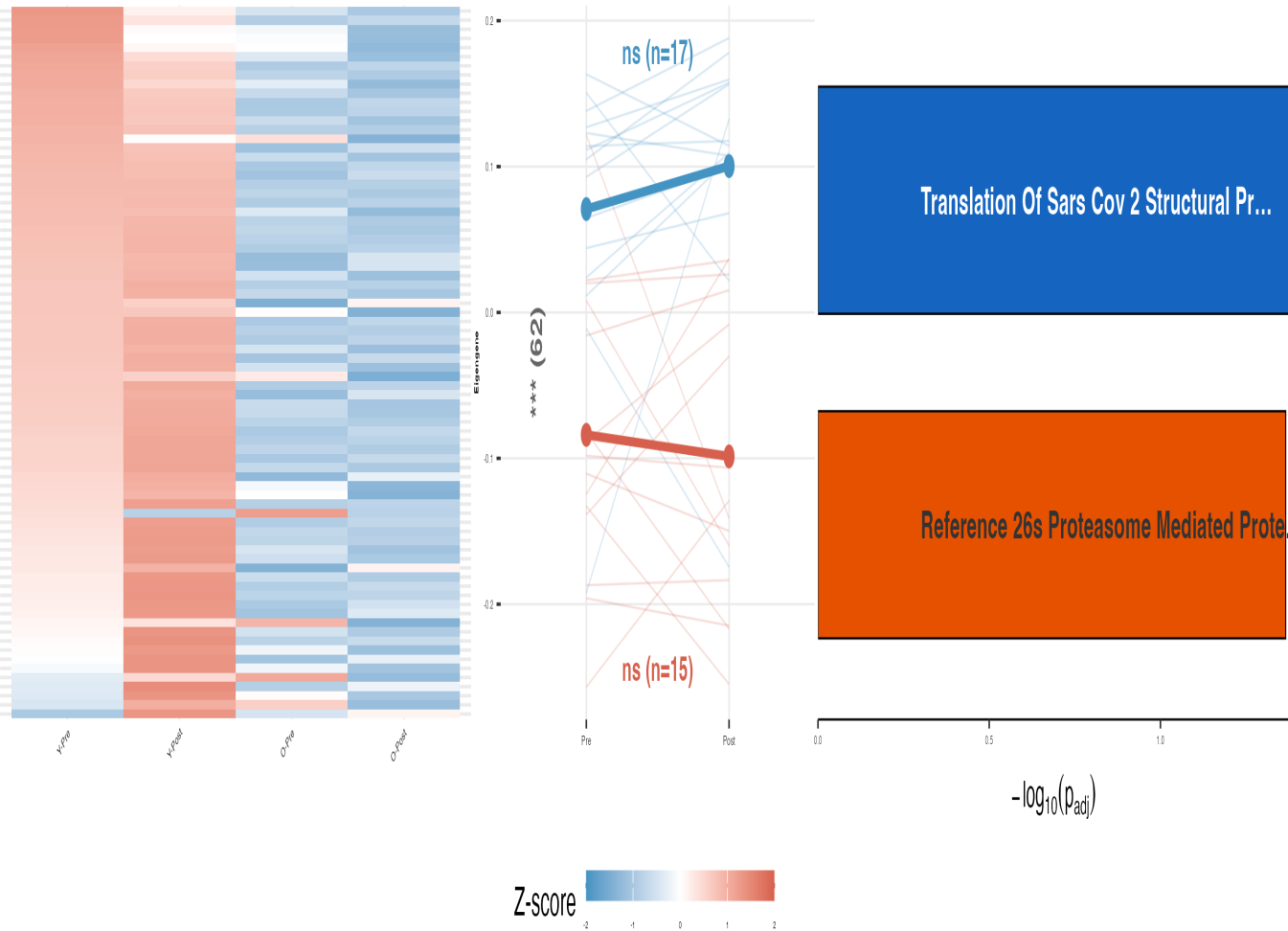
*FDR < 0.5 **FDR < 0.01 ***FDR < 0.001 (all LMM Bonferroni)

E

M9: Translation Initiation

Aging: $r = +0.53$

M9: Translation Initiation (n=78)



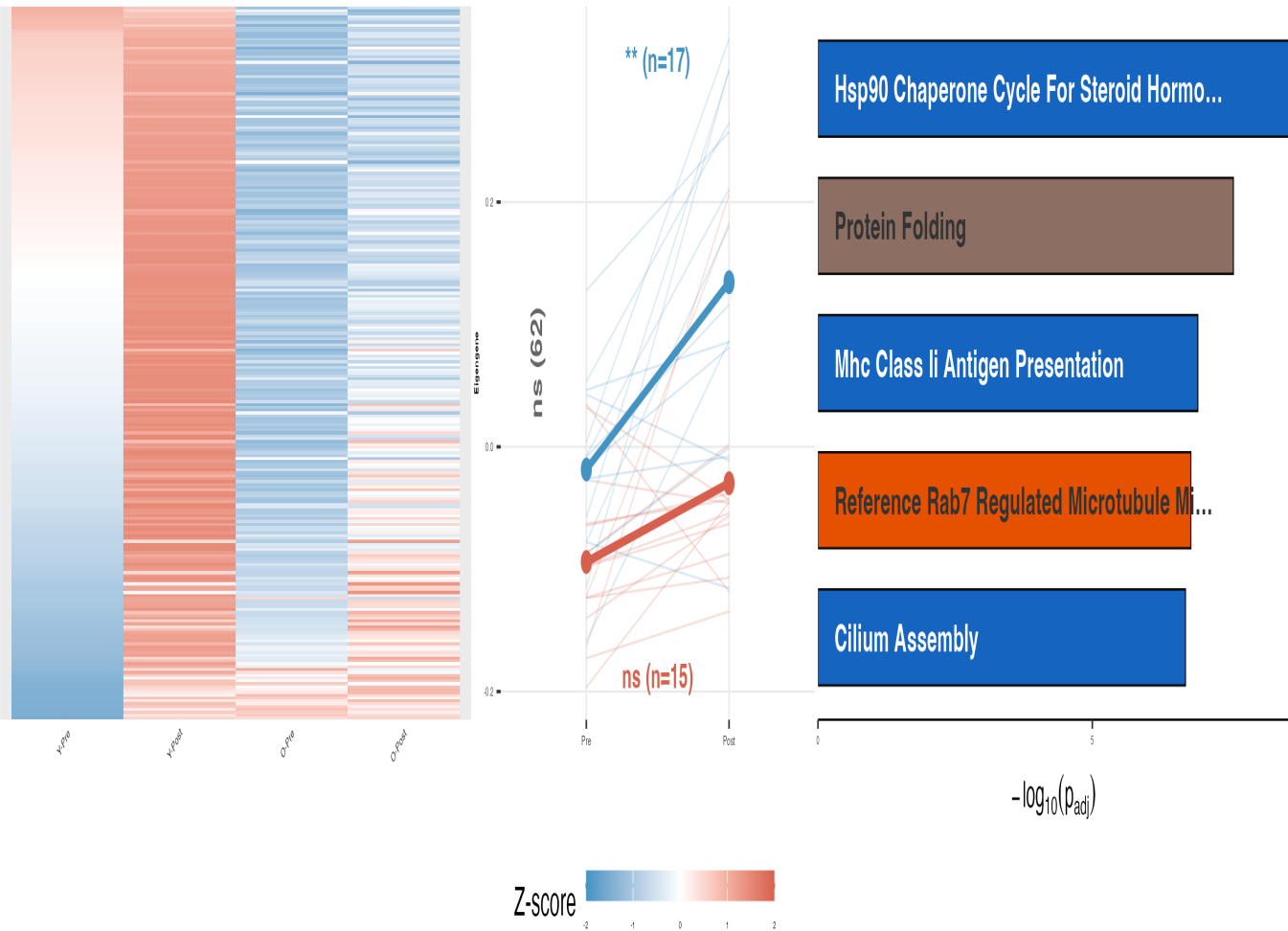
*FDR < 0.5 **FDR < 0.01 ***FDR < 0.001 (all LMM Bonferroni)

B

M2: Cell Cycle/Proteostasis

Training Young: $r = +0.62$

M2: Cell Cycle/Proteostasis (n=250)



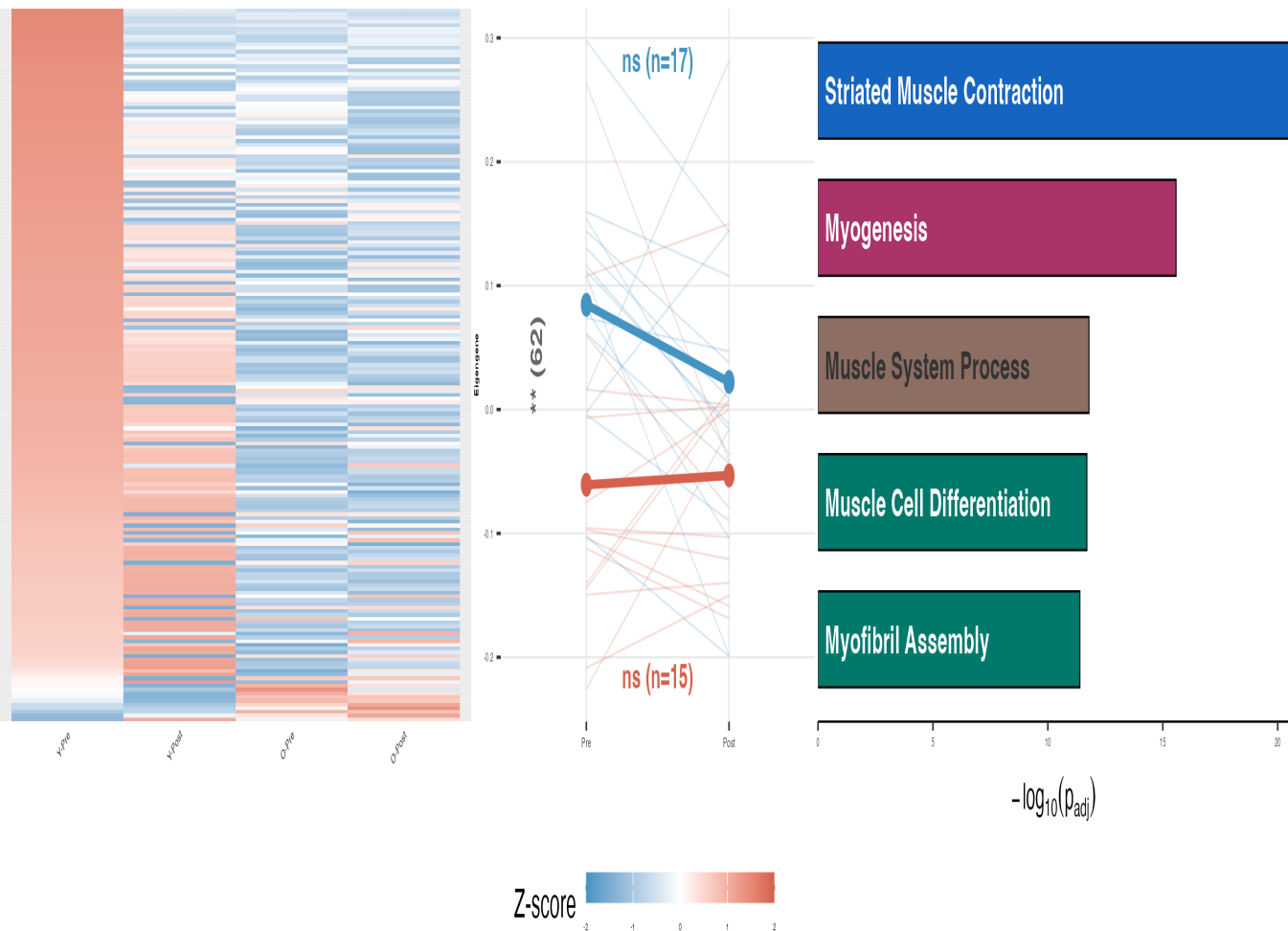
*FDR < 0.5 **FDR < 0.01 ***FDR < 0.001 (all LMM Bonferroni)

D

M3: Redox/Glycolysis

Aging: $r = +0.44$

M3: Redox/Glycolysis (n=191)

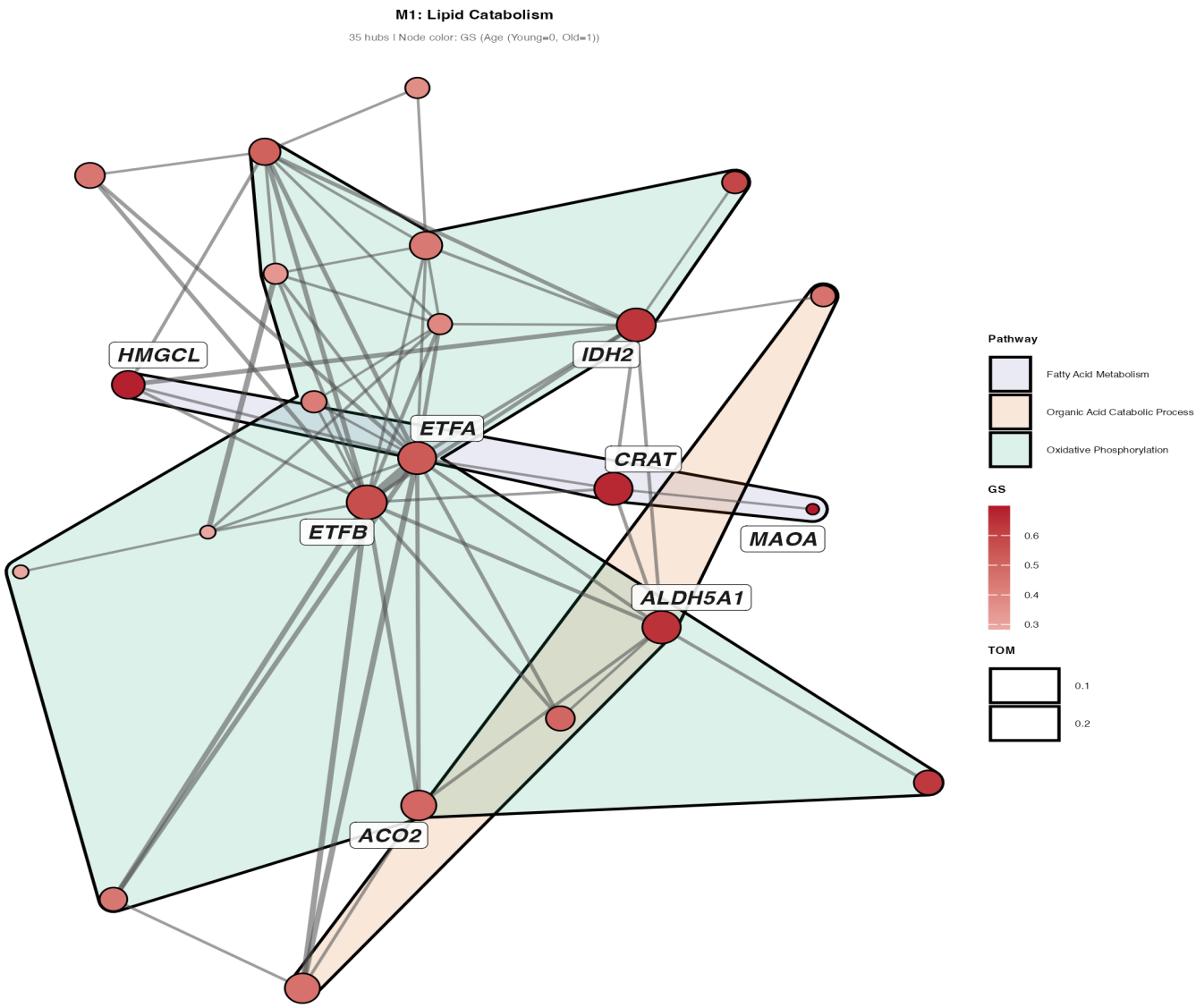


*FDR < 0.5 **FDR < 0.01 ***FDR < 0.001 (all LMM Bonferroni)

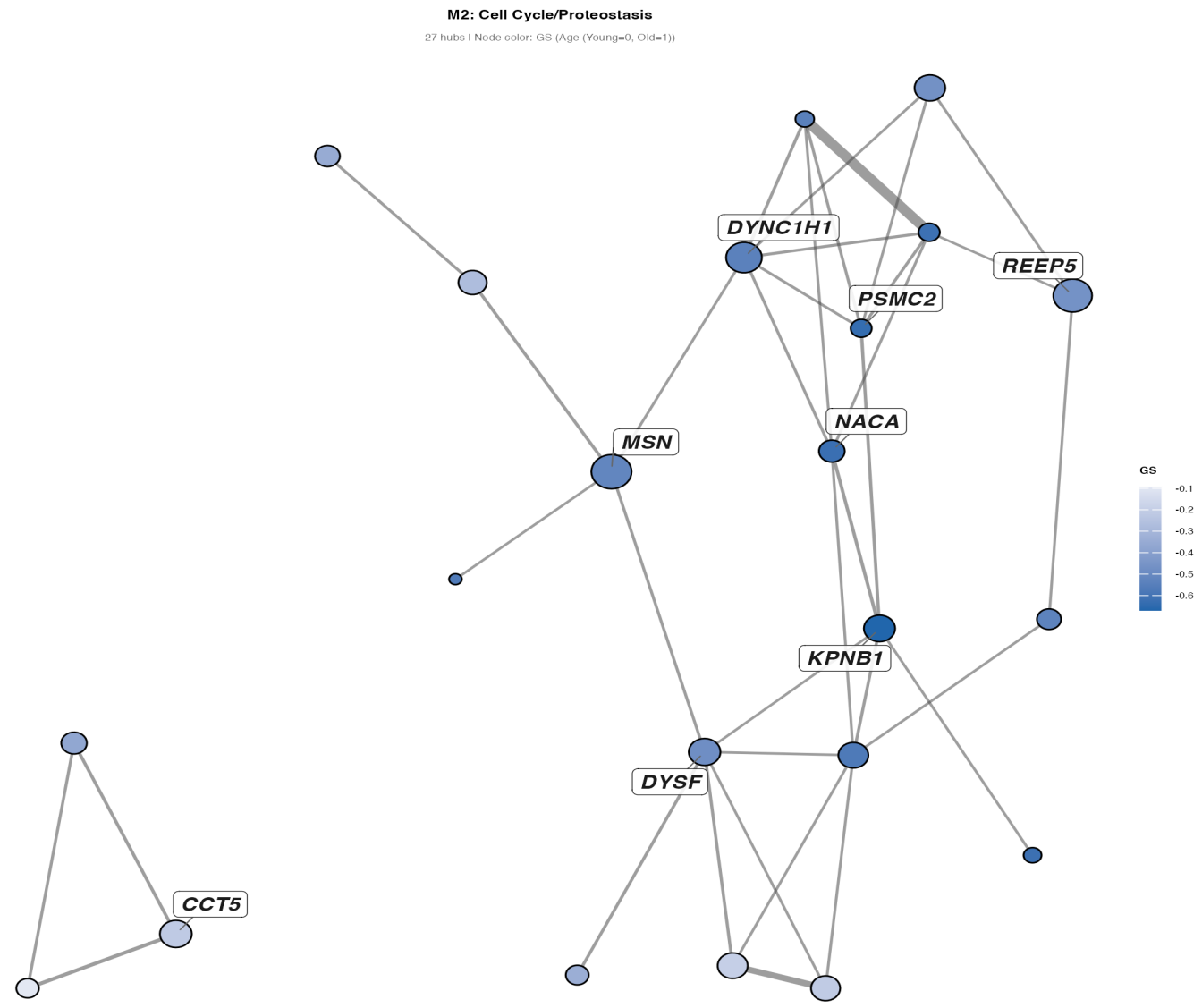
S7d – Hub Protein Networks (Key Modules)

A: Turquoise | B: Blue | C: Green | D: Brown | E: Magenta | F: Hub chord diagram

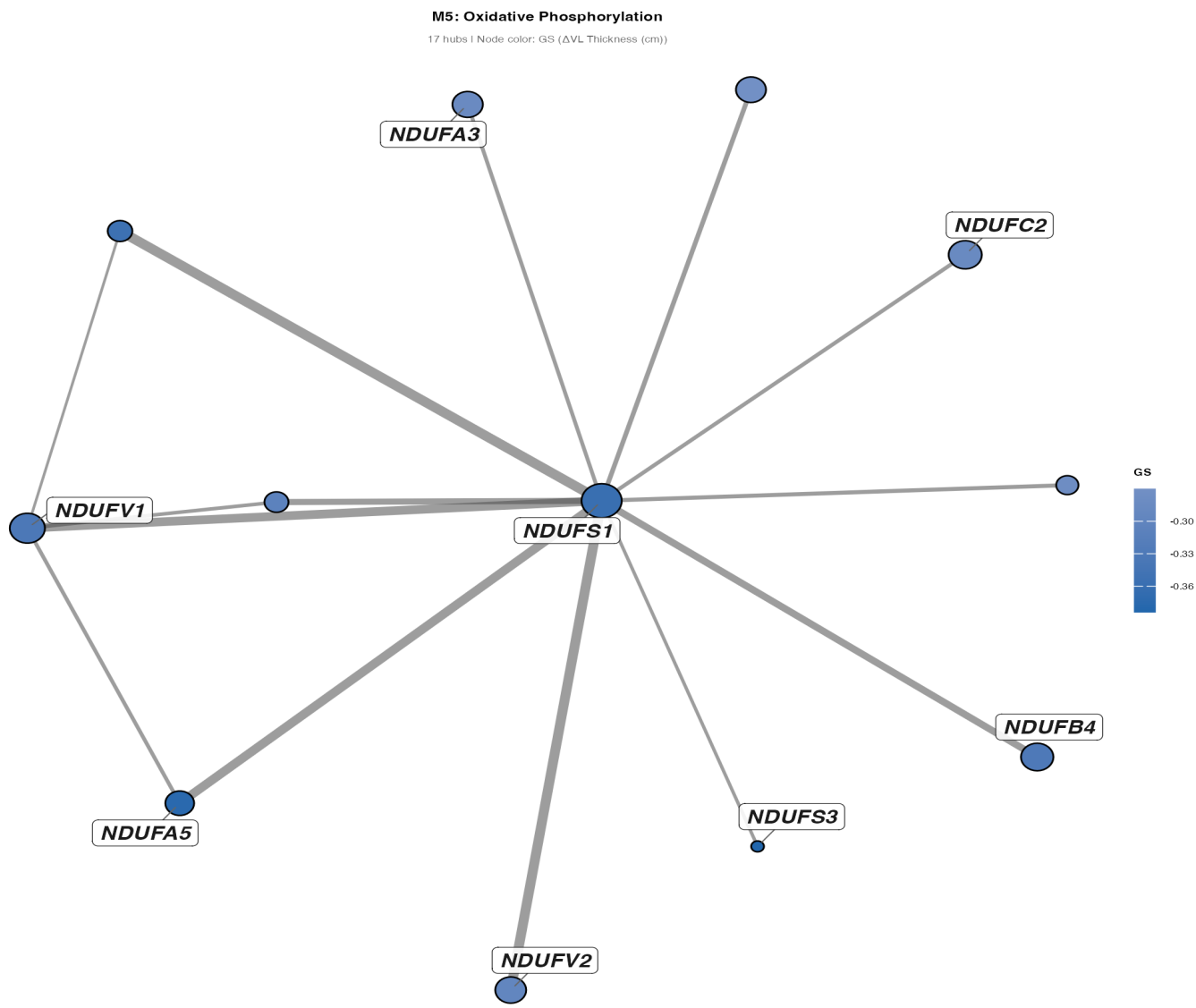
A



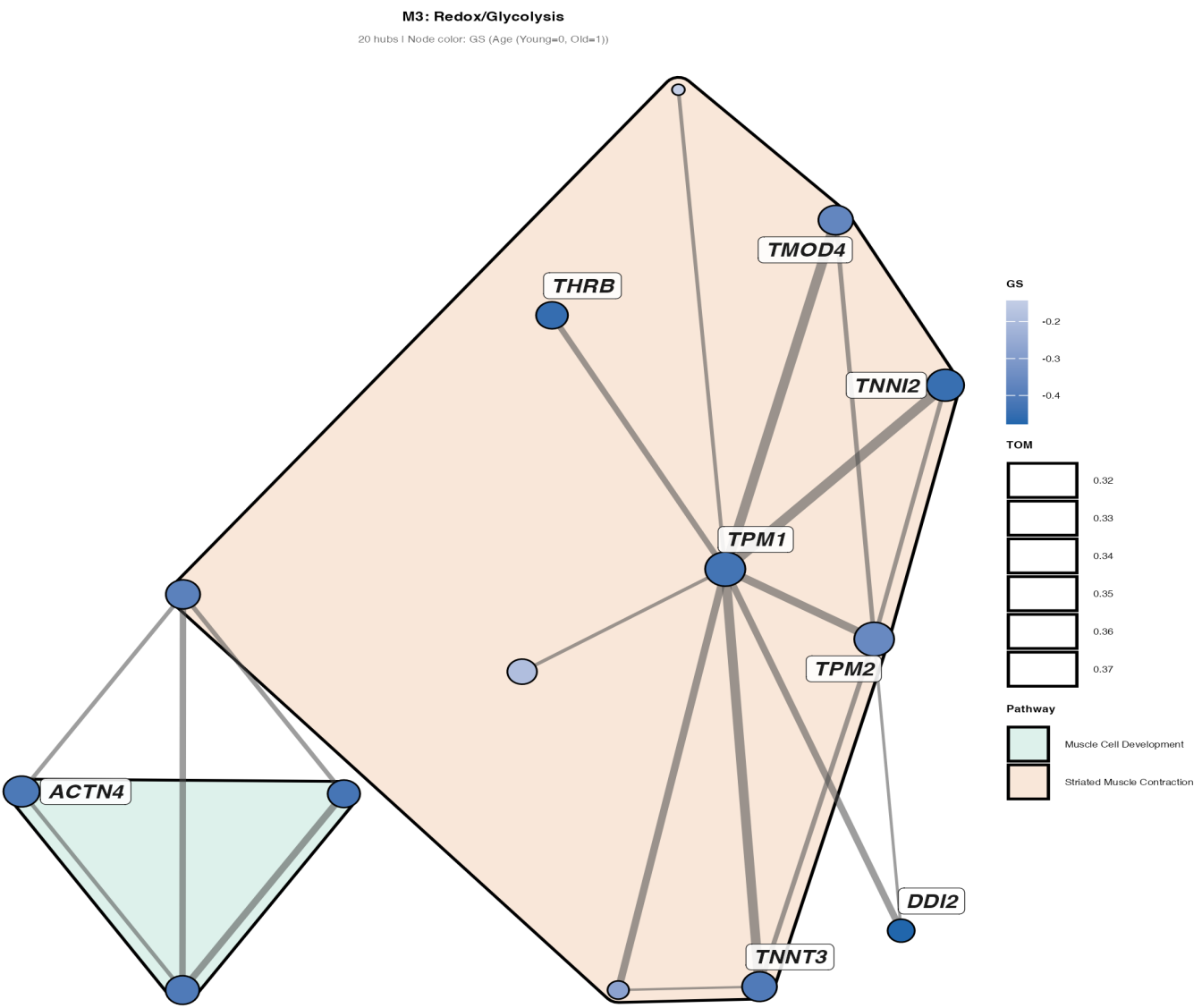
B



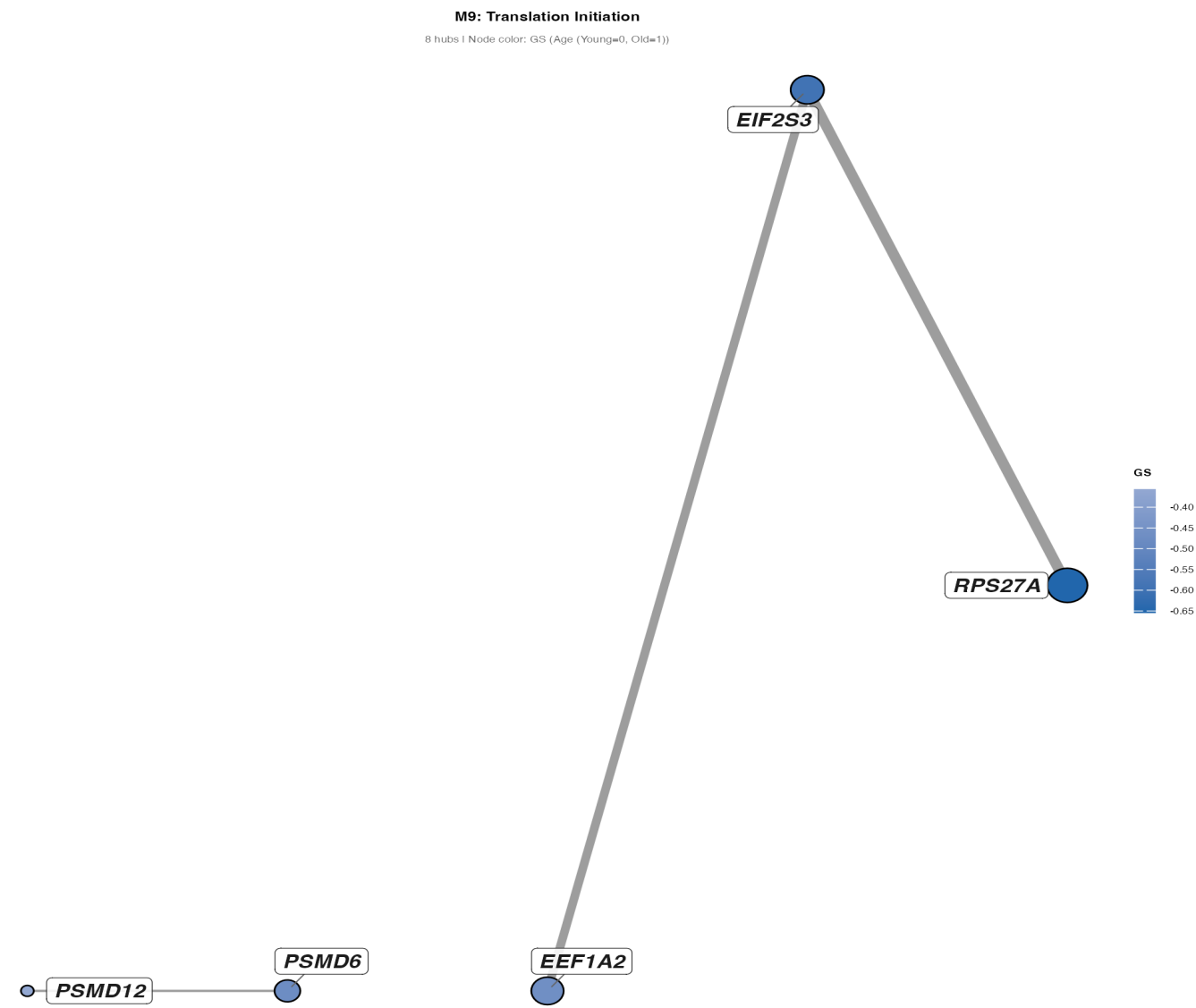
C



D



E



F

